

Draw.io Agent Degraded-Mode Runbook

DRR-2026-09 | retrieval, OCR and canvas recovery

Scope

Runbook DRR-2026-09 defines user-visible behavior when a dependency fails during material-grounded diagram creation. It covers vector search, object storage, OCR, visual verification and canvas persistence. DRR-2026-90 is a load-test sheet.

Invariant

Failure never expands authorization, changes a pinned version or converts an unverified fact into a verified one. The user sees which capability is unavailable and which operations remain safe.

Severity

SEV-1 means possible cross-scope disclosure or corrupted saved canvas. SEV-2 means a material-backed task cannot complete without data loss. SEV-3 means a retryable delay with preserved work.

No secrets

This fixture contains no live host, API key, on-call phone number or customer identifier.

1. Failure matrix

Dependency, state and allowed action

Matrix rule

The allowed action is conservative. A capability may stay available only when it does not depend on the failed evidence path.

Failure	User state	Allowed	Blocked	Severity
Pinecone	retrieval degraded	manual drawing	material search	SEV-2
Object store	hydration failed	edit saved XML	open evidence	SEV-2
OCR	scan unavailable	native PDF	scan claims	SEV-3
Visual verifier	visual pending	text claims	pixel relations	SEV-3
Canvas save	save blocked	download draft	publish	SEV-1

Matrix result

During a Pinecone failure, manual drawing remains available but material search is blocked. A canvas-save failure is SEV-1 because publishing could lose or corrupt user work.

2. Detection and containment route

Raster-only incident flow

Figure scope

Figure 2 carries the containment sequence. The relation following FREEZE MATERIAL TASKS is not repeated in native text.

Containment

In-flight material tasks keep their evidence manifest for diagnosis, but no partial grounded answer is published. Ordinary local shape edits may continue when canvas persistence is healthy.

DEGRADED-MODE CONTAINMENT



3. Recovery verification

Probe order, deadlines and false positives

Probe order

First verify tenant and chartbook filters, then version selection, then evidence hydration, and finally diagram generation. A successful unfiltered query is not a valid recovery probe.

Search recovery

Pinecone returns to normal only after 12 consecutive scoped probes succeed across three test tenants. The nearby value 21 belongs to a capacity drill and is not the recovery threshold.

OCR recovery

OCR recovery requires anchor recall of at least 0.95 on the pinned scan suite and no cross-page substitution. Character error rate alone cannot prove row or arrow correctness.

Canvas recovery

A saved draft is reopened, edited and saved again before publish is enabled. The test must preserve stable cell identifiers and citation bindings.

4. User messaging and retry policy

Honest degraded states

Message

The user message names the unavailable capability and states whether work was preserved. It never says 'no evidence exists' when retrieval did not execute.

Automatic retry

SEV-3 operations retry twice with bounded backoff. SEV-1 and SEV-2 require a verified dependency state before retrying material work.

Fallback

Manual drawing and edits to already loaded shapes may continue if they do not add material-backed facts. Model knowledge is not an invisible substitute for failed retrieval.

Cancellation

The user may cancel a queued retry without deleting the saved canvas or evidence audit record.

Status clear

The degraded banner clears only after the same scoped operation that failed has passed verification.

5. Approval and post-incident evidence

Exit criteria and unsupported requests

Approval

Runbook version 1.0 was approved on 21 December 2026 by the Reliability Council. A draft allowed five scoped probes; it was rejected in favor of twelve consecutive probes.

Exit criteria

Closure requires the incident timeline, affected capability, scope-verification results, recovery probes, user-message sample and confirmation that temporary vectors were deleted.

Owner

The Incident Commander approves service restoration. The Diagram team validates canvas integrity, while the Retrieval team validates scoped search. Neither team can approve its own authorization exception.

Unsupported

No live API key, endpoint, customer record or private on-call number appears. Those requests require abstention.